AQ8.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

1 point
Consider two vectors $a = (2, 0, 3, 0, 8)$, $b = (3, 2, -2, 4, 0)$ in R^5. Choose the set of correct options.

- [ ] a and b are orthogonal.
- [ ] a and b are not orthogonal.
- [ ] $(a - b) \cdot a = 0.$
- [ ] $(a - b) \cdot a = 77.$

No, the answer is incorrect.
Score: 0
Accepted Answers:
a and b are orthogonal.
$(a - b) \cdot a = 77.$

1 point
Choose the set of correct statements.

- [ ] In an orthogonal set, the norms of all the vectors are equal.
- [ ] In an orthogonal set, the vectors are linearly independent.
- [ ] In an orthogonal set, the vectors are linearly dependent.
- [ ] If the columns of an $n \times n$ coefficient matrix A comprises of the individual vectors of an orthogonal set in R^n, then there must be a unique solution to the system $AX = b$, where X, b are $n \times 1$ vectors.
- [ ] If the columns of an $n \times n$ coefficient matrix A comprises of the individual vectors of an orthogonal set in R^n, then there are no solutions to the system $AX = b$, where X, b are $n \times 1$ vectors.
- [ ] The determinant of a square matrix formed by a set of orthogonal vectors in R^n is zero.
- [ ] A set of n vectors can never form an orthogonal basis in R^{n-1}.

No, the answer is incorrect.
Score: 0
Accepted Answers:
In an orthogonal set, the vectors are linearly independent.
If the columns of an $n \times n$ coefficient matrix A comprises of the individual vectors of an orthogonal set in R^n, then there must be a unique solution to the system $AX = b$, where X, b are $n \times 1$ vectors.
A set of n vectors can never form an orthogonal basis in R^{n-1}.

1 point
Which of the following is an orthogonal basis of the given vector spaces with respect to the sta inner product (dot product)?

- [ ]

$\{(1,0),(0,1)\}$ is an orthogonal basis of R^2.

☐

$\{(1,0,0),(0,1,0),(0,0,1)\}$ is an orthogonal basis of R^3.

☐

$\{(3,4),(4,-3),(2,-3)\}$ is an orthogonal basis of R^2.

☐

$\{(2,1,-1),(-1,1,-1),(3,-3,3)\}$ is an orthogonal basis of R^3.

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\{(1,0),(0,1)\}$ is an orthogonal basis of R^2.

$\{(1,0,0),(0,1,0),(0,0,1)\}$ is an orthogonal basis of R^3.

Level 2

1 point

Find a vector in R^4 that is orthogonal to the subspace spanned by $(1,1,0,0)$ and $(0,1,1,0)$ with respect to the dot product as the inner product.

☐ (1,-1,1,0)
☐ (2,3,4,5)
☐ (1,-1,1,1)
☐ (1,1,1,0)

No, the answer is incorrect.
Score: 0
Accepted Answers:

(1,-1,1,0)
(1,-1,1,1)

1 point

Which of the following are orthogonal to the vector $(1,2)$ in R^2 with respect to the inner product $< v, w > = v_1 w_1 - v_1 w_2 - v_2 w_1 + 4 v_2 w_2$?

☐

$(-7,1)$

☐

$(6,-1)$

☐

$(7,1)$

☐

$(-9,1)$

☐

$(14,2)$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$(7,1)$

$(14,2)$

Consider the system of linear equations:

$$x_1 - 2x_2 + 3x_3 = 1$$

$2x_1 + x_2 = 5$2x1 + x2 = 5

$3x_1 - 6x_2 - 5x_3 = 9$.3x1 − 6x2 − 5x3 = 9.

Number of solutions of the above system is

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point
1 point
Let AA be the coefficient matrix of the system of linear equations:

$x_1 - 2x_2 + 3x_3 = 1$x1 − 2x2 + 3x3 = 1

$2x_1 + x_2 = 5$2x1 + x2 = 5

$3x_1 - 6x_2 - 5x_3 = 9$.3x1 − 6x2 − 5x3 = 9.

Which one of the following is true about the matrix AA^T AAT ?
- [ ] A scalar matrix.
- [ ] The identity matrix.
- [ ] A diagonal matrix.
- [ ] A lower triangular matrix.
- [ ] An upper triangular matrix.
- [ ] None of the above.

No, the answer is incorrect.
Score: 0
Accepted Answers:
A diagonal matrix.
A lower triangular matrix.
An upper triangular matrix.

Check Answers
Your score is: 0/7